

A second order unstructured finite volume method for incompressible Navier-Stokes equations using face based constrained least-squares procedure

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Abstract

In this paper, we discuss the use of face based constrained least-squares reconstruction for a second-order cell-centered finite volume discretization of incompressible flows on unstructured grids. In a cell-centered finite volume method, although variables at cell centers are updated, calculation of fluxes and application of boundary conditions is performed at faces of the mesh.

1. Methodology

1.1. Constrained Least Squares

The constrained least squares is [1] used for reconstruction [6] to solve incompressible Navier-Stokes [3]

1.2. Reconstruction for interior faces

The constrained least-squares system for reconstruction on an interior face of the mesh is

$$\begin{bmatrix} 2(N_c - 2) & 2 \sum \Delta x_i & 2 \sum \Delta y_i & 1 & 1 \\ 2 \sum \Delta x_i & 2 \sum \Delta x_i^2 & 2 \sum \Delta x_i \Delta y_i & \Delta x_1 & \Delta x_2 \\ 2 \sum \Delta y_i & 2 \sum \Delta x_i \Delta y_i & 2 \sum \Delta y_i^2 & \Delta y_1 & \Delta y_2 \\ 1 & \Delta x_1 & \Delta y_1 & 0 & 0 \\ 1 & \Delta x_2 & \Delta y_2 & 0 & 0 \end{bmatrix} \begin{bmatrix} \phi_0 \\ \phi_x \\ \phi_y \\ \hat{z}_1 \\ \hat{z}_2 \end{bmatrix} = \begin{bmatrix} 2 \sum \phi_i \\ 2 \sum \Delta x_i \phi_i \\ 2 \sum \Delta y_i \phi_i \\ \phi_1 \\ \phi_2 \end{bmatrix} \quad (1)$$

1.3. Reconstruction for faces on Dirichlet boundaries

$$\begin{bmatrix} 2(N_c - 2) & 2 \sum \Delta x_i & 2 \sum \Delta y_i & 1 & 1 \\ 2 \sum \Delta x_i & 2 \sum \Delta x_i^2 & 2 \sum \Delta x_i \Delta y_i & \Delta x_1 & 0 \\ 2 \sum \Delta y_i & 2 \sum \Delta x_i \Delta y_i & 2 \sum \Delta y_i^2 & \Delta y_1 & 0 \\ 1 & \Delta x_1 & \Delta y_1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \phi_0 \\ \phi_x \\ \phi_y \\ \hat{z}_1 \\ \hat{z}_2 \end{bmatrix} = \begin{bmatrix} 2 \sum \phi_i \\ 2 \sum \Delta x_i \phi_i \\ 2 \sum \Delta y_i \phi_i \\ \phi_1 \\ \phi_b \end{bmatrix} \quad (2)$$

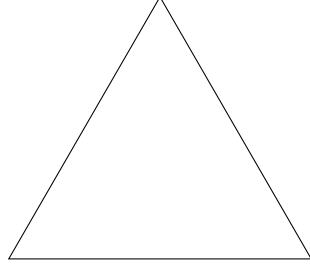


Figure 1: Arrangement of variables in a cell

1.4. Reconstruction for faces on Neumann boundaries

$$\begin{bmatrix} 2(N_c - 2) & 2 \sum \Delta x_i & 2 \sum \Delta y_i & 1 & 0 \\ 2 \sum \Delta x_i & 2 \sum \Delta x_i^2 & 2 \sum \Delta x_i \Delta y_i & \Delta x_1 & n_x \\ 2 \sum \Delta y_i & 2 \sum \Delta x_i \Delta y_i & 2 \sum \Delta y_i^2 & \Delta y_1 & n_y \\ 1 & \Delta x_1 & \Delta y_1 & 0 & 0 \\ 0 & n_x & n_y & 0 & 0 \end{bmatrix} \begin{bmatrix} \phi_0 \\ \phi_x \\ \phi_y \\ \hat{z}_1 \\ \hat{z}_2 \end{bmatrix} = \begin{bmatrix} 2 \sum \phi_i \\ 2 \sum \Delta x_i \phi_i \\ 2 \sum \Delta y_i \phi_i \\ \phi_1 \\ \phi_b \end{bmatrix} \quad (3)$$

All summations in the matrix above are from $i = 3$ to N_c

2. Results

2.1. Potential flow around a circular cylinder

Potential flow over a cylinder models inviscid, incompressible, and irrotational fluid flowing around a circular body. Since the velocity field is irrotational, it can be expressed in terms of a potential function ϕ [7], as $\mathbf{u} = \nabla \phi$. Further, the divergence free condition on velocity, $\nabla \cdot \mathbf{u}$, leads to a Poisson equation, $\nabla^2 \phi = 0$, for the potential function. For potential flow past cylinder, the exact solution for radial and tangential velocity components u_r and u_θ , is given in polar coordinates (r, θ) by

$$\phi = \left(r + \frac{R^2}{r} \right) U \cos \theta \quad (4a)$$

$$u_r = \frac{\partial \phi}{\partial r} = \left(1 - \frac{R^2}{r^2} \right) U \cos \theta \quad (4b)$$

$$u_\theta = \frac{1}{r} \frac{\partial \phi}{\partial \theta} = - \left(1 + \frac{R^2}{r^2} \right) U \sin \theta \quad (4c)$$

For our test case, we use free stream velocity $U = 1$. The computational domain is a square $\Omega = [-5, 5]^2$ at the center of which a circular cylinder of radius $R = 1$ is placed. On surface of the cylinder, no flow through condition, $\mathbf{u} \cdot \mathbf{n} = 0$, translates to a homogeneous Neumann condition, $\nabla \phi \cdot \mathbf{n} = 0$, for the potential function.

On boundaries of the outer square domain, we impose exact solution provided by Equation 4. A schematic diagram for the problem showing the geometry and boundary conditions is shown in Figure 2.

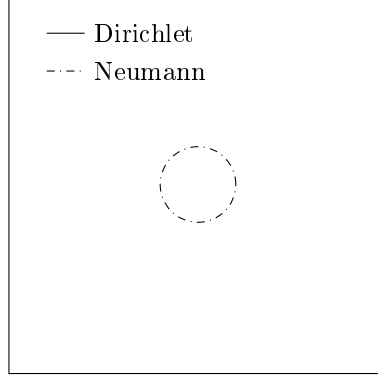


Figure 2: Schematic diagram showing geometry and boundary conditions for potential past a circular cylinder test case.

Table 1 shows convergence rates on a sequence of meshes with characteristic size $h = 2\pi/64$ through $h = 2\pi/1024$. With subsequent refinement, the desired second order convergence rate is achieved in both L_2 and L_∞ norms.

h	L_2 error	L_2 convergence	L_∞ Error	L_∞ convergence
$2\pi/64$	1.201×10^{-3}		5.728×10^{-3}	
$2\pi/128$	2.920×10^{-4}	2.04	1.783×10^{-3}	1.68
$2\pi/256$	6.989×10^{-5}	2.06	4.808×10^{-4}	1.89
$2\pi/512$	1.697×10^{-5}	2.04	1.235×10^{-4}	1.96
$2\pi/1024$	4.176×10^{-6}	2.02	3.120×10^{-5}	1.98

Table 1: L_2 and L_∞ errors for potential flow around a cylinder test, as a function of characteristic mesh size h . In the present case, h is calculated by dividing perimeter of the cylinder by number of grid points on it.

2.2. Kovasznay flow solution

The first test case for incompressible Navier-Stokes equations is the Kovasznay analytical solution for laminar flow [4]. This test case provides a non-trivial exact solution for steady-state Navier-Stokes equations, and is given by

$$u = 1 - e^{\lambda x} \cos(2\pi y) \quad (5a)$$

$$v = \frac{\lambda}{2\pi} e^{\lambda x} \sin(2\pi y) \quad (5b)$$

$$p = \frac{1}{2}(1 - e^{2\lambda x}) \quad (5c)$$

In the above equations $\lambda = \frac{1}{2\nu} - \sqrt{\frac{1}{4\nu^2} + 4\pi^2}$.

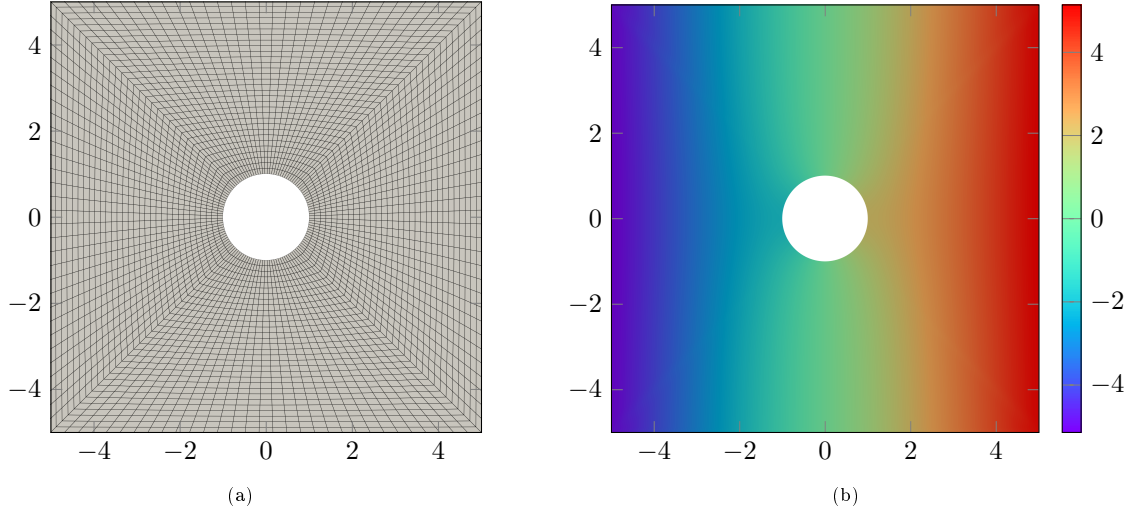


Figure 3: Solution for potential flow past a circular cylinder. (a) grid with characteristic mesh space $h = \frac{2\pi}{128}$ and (b) potential function ϕ in the domain.

2.3. Lid Driven Cavity

2.4. Unsteady flow around cylinder in a channel

3. Conclusions

Acknowledgments

The Library of Iterative solvers for Linear Systems (LIS) [5] is used to solve the system. The least-squares systems are solved using [2]

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